Computing the Aligned Deformation Gradient $\mathbf{F}_{\text{aligned}}$

Removing Arbitrary Mesh-Triangulation Orientations in First Piola–Kirchhoff and Projected Stress Post-Processing

MTM Lattice-Triangulation Documentation

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Abstract

In a 2D triangulated crystal mesh, each triangular element's raw deformation gradient $\mathbf{F}$ is defined relative to a single fixed canonical right-angled reference triangle. Because a unit cell consists of both lower-left ($\triangle$) and upper-right ($\triangledown$) triangles, and because CGAL Delaunay triangulation assigns counterclockwise vertex indices (v_0, v_1, v_2) starting from an arbitrary corner, the raw deformation gradient $\mathbf{F}$ contains an arbitrary discrete reference-space transformation $\mathbf{N}^{-1} \in SL(2,\mathbb{Z})$ on the right. While the metric tensor $\mathbf{C} = \mathbf{F}^{\mathsf{T}}\mathbf{F}$ (after Lagrange reduction), the strain energy $W(\mathbf{C}_{\text{red}})$, and the Cauchy stress $\boldsymbol{\sigma}$ are strictly invariant under $\mathbf{N} \in SL(2,\mathbb{Z})$, the First Piola–Kirchhoff stress $\mathbf{P} = \partial W/\partial \mathbf{F}$ and the Projected Stress $\tau = \mathbf{m}\cdot(\mathbf{P}\mathbf{n})$ Carry an explicit reference-space index and therefore flip sign or swap columns between neighboring triangles. This document explains how $\mathbf{F}_{\text{aligned}} = \mathbf{F}\mathbf{N}$ is computed and how the entire post-processing stress pipeline follows directly from $\mathbf{F}_{\text{aligned}}$ without altering any downstream formulas.

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1 Why the Raw Deformation Gradient F Depends on Triangle Labelling

Let a triangular element have current deformed vertex positions $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^2$ ordered counterclockwise ($\det[\mathbf{x}_1 - \mathbf{x}_0,\ \mathbf{x}_2 - \mathbf{x}_0] > 0$). In the MTM code (`ElementTriangle2D::calculate_deformation_gradie` every element is mapped from the **same canonical right-angled reference triangle** with edge matrix

$$\mathbf{X}_{\text{ref}} = \begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix}, \qquad \mathbf{X}_{\text{ref}}^{-1} = \frac{1}{a_0}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1}$$

Denoting the two deformed edge vectors emanating from vertex v_0 by

$$\mathbf{e}_1 = \mathbf{x}_1 - \mathbf{x}_0, \qquad \mathbf{e}_2 = \mathbf{x}_2 - \mathbf{x}_0, \qquad \mathbf{E} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{bmatrix} \in \mathbb{R}^{2\times 2}, \tag{2}$$

the raw element deformation gradient is computed as

$$\mathbf{F} = \frac{1}{a_0}\,\mathbf{E}\,\mathbf{X}_{\text{ref}}^{-1} \qquad \text{(equal to } \mathbf{E} \text{ when normalized by } a_0 = 1). \tag{3}$$

Two purely geometric ambiguities enter Eq. (3) before any physical elastic deformation occurs:

1.1 Ambiguity 1: The 4 Discrete 90° Half-Cell Orientations $(\mathbf{R}_0, \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3)$

In a finite-element mesh with dynamic Delaunay remeshing, triangles are destroyed and re-created during plastic avalanches, so individual triangles do not store their own initial node coordinates. Instead, every triangle in the mesh is compared against the **single bottom-left template triangle** $p_1 = (0,0),\ p_2 = (a_0, 0),\ p_3 = (0, a_0)$ via the fixed shape-derivative matrix `dndx`.

However, a unit square $[0,1] \times [0,1]$ has four corners and can be split along either diagonal (\ or /) into four possible right-angled half-cell triangles whose 90° corner sits at $(0,0)$, $(1,0)$, $(1,1)$, or $(0,1)$. Even when the origin v_0 is placed directly at the 90° corner of each triangle, walking counterclockwise produces four different raw deformation gradients rotated by $0°, 90°, 180°, 270°$:

1. **Bottom-left triangle** (90° corner at $v_0 = (0,0) \to (1,0) \to (0,1)$, rotation 0°):

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{R}_0. \tag{4}$$

2. **Bottom-right triangle** (90° corner at $v_0 = (1,0) \to (1,1) \to (0,0)$, rotation 90°):

$$\mathbf{e}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \mathbf{R}_1. \tag{5}$$

3. **Top-right triangle** (90° corner at $v_0 = (1,1) \to (0,1) \to (1,0)$, rotation 180°):

$$\mathbf{e}_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \mathbf{R}_2. \tag{6}$$

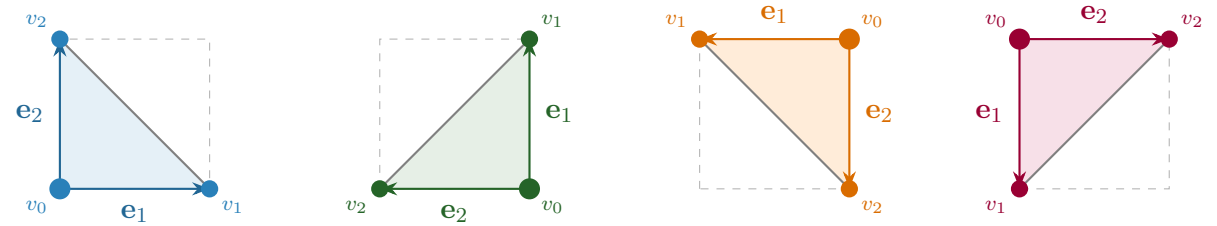
4. **Top-left triangle** (90° corner at $v_0 = (0,1) \to (0,0) \to (1,1)$, rotation 270°):

$$\mathbf{e}_1 = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathbf{R}_3. \tag{7}$$

Right-multiplying $\mathbf{F}$ by $\mathbf{R}_r^{\mathsf{T}}$ ($r \in \{0,1,2,3\}$) rotates the local reference axes of the triangle by $-0°, -90°, -180°, -270°$, bringing all four half-square triangles back to $\mathbf{F}\mathbf{R}_r^{\mathsf{T}} = +\mathbf{I}$.

1.2 Ambiguity 2: The 3 Cyclic Counterclockwise Vertex Shifts in CGAL $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2)$

Now consider a **single fixed triangle** in space—for example, the bottom-left triangle with corners at $(0,0)$, $(1,0)$, and $(0,1)$. When CGAL builds or remeshes the Delaunay triangulation, any of the triangle's three corners may be listed first as v_0, provided (v_0, v_1, v_2) remains counterclockwise. Cycling which corner is chosen as the origin v_0 transforms the edge matrix $\mathbf{E} = [\mathbf{e}_1, \mathbf{e}_2]$ by right-multiplication with a unimodular matrix $\mathbf{M}_p \in SL(2, \mathbb{Z})$:



0° ($\mathbf{R}_0$): Bottom-Left — $v_0 = (0,0)$ — $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

90° ($\mathbf{R}_1$): Bottom-Right — $v_0 = (1,0)$ — $\mathbf{F} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

180° ($\mathbf{R}_2$): Top-Right — $v_0 = (1,1)$ — $\mathbf{F} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

270° ($\mathbf{R}_3$): Top-Left — $v_0 = (0,1)$ — $\mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Figure 1: **What the 4 discrete rotations $\mathbf{R}_r$ do:** A unit square $[0,1] \times [0,1]$ (dashed outline) can contain four different right-angled triangle orientations depending on which half of the square the triangle occupies and which diagonal (\ or /) splits the cell. Right-multiplying by $\mathbf{R}_r^{\mathsf{T}}$ rotates all four orientations back to $+\mathbf{I}$.

1. **No shift ($\mathbf{M}_0$)**, origin at $v_0 = (0,0) \to v_1 = (1,0) \to v_2 = (0,1)$:

$$[\mathbf{e}_1,\ \mathbf{e}_2] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\mathbf{M}_0}. \tag{8}$$

2. **One cyclic shift ($\mathbf{M}_1$)**, origin shifted to $v_0' = (1,0) \to v_1' = (0,1) \to v_2' = (0,0)$: new edges are $\mathbf{e}_1' = \mathbf{x}_2 - \mathbf{x}_1 = -\mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{e}_2' = \mathbf{x}_0 - \mathbf{x}_1 = -\mathbf{e}_1$:

$$[\mathbf{e}_1',\ \mathbf{e}_2'] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}}_{\mathbf{M}_1}. \tag{9}$$

3. **Two cyclic shifts ($\mathbf{M}_2$)**, origin shifted to $v_0'' = (0,1) \to v_1'' = (0,0) \to v_2'' = (1,0)$: new edges are $\mathbf{e}_1'' = \mathbf{x}_0 - \mathbf{x}_2 = -\mathbf{e}_2$ and $\mathbf{e}_2'' = \mathbf{x}_1 - \mathbf{x}_2 = \mathbf{e}_1 - \mathbf{e}_2$:

$$[\mathbf{e}_1'',\ \mathbf{e}_2''] = [\mathbf{e}_1,\ \mathbf{e}_2] \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}}_{\mathbf{M}_2}. \tag{10}$$

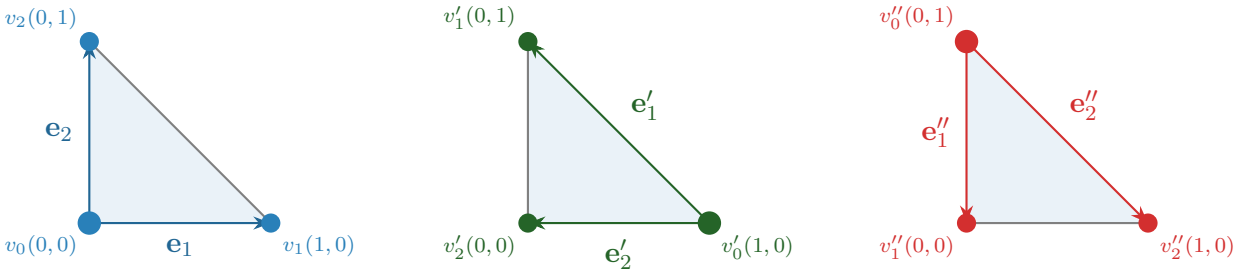
Note that $\det \mathbf{M}_0 = \det \mathbf{M}_1 = \det \mathbf{M}_2 = +1$, and $\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2 \in SL(2,\mathbb{Z})$ form a cyclic group of order 3 ($\mathbf{M}_1^2 = \mathbf{M}_2$, $\mathbf{M}_1^3 = \mathbf{I}$, so $\mathbf{M}_1^{-1} = \mathbf{M}_2$ and $\mathbf{M}_2^{-1} = \mathbf{M}_1$).

2 Single Unified Formula for Computing $\mathbf{F}_{\text{aligned}}$

Because a crystal in the Zanzotto potential can contain both square and triangular (hexagonal) phases simultaneously (as well as heavily sheared dislocation cores), we use a **single unified formula** for every element without branching on lattice type.

In the Lagrange fundamental domain $0 \le 2\,C_{12} \le C_{11} \le C_{22}$:

- The **square phase** sits at the left boundary $F_{12} = 0$: $\mathbf{F}_{\text{sq}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.
- The **triangular (hexagonal) phase** sits at the right boundary $F_{12} = +\frac{1}{2}a_0$: $\mathbf{F}_{\text{tri}} = a_0 \begin{pmatrix} 1 & +1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}$.



No shift ($\mathbf{M}_0$)
Origin at $v_0 = (0,0)$
$\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

1 cyclic shift ($\mathbf{M}_1$)
Origin at $v_0' = (1,0)$
$\mathbf{F} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$

2 cyclic shifts ($\mathbf{M}_2$)
Origin at $v_0'' = (0,1)$
$\mathbf{F} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$

Figure 2: **What the 3 cyclic shifts $\mathbf{M}_p$ do:** All three panels show the **exact same physical triangle** with vertices at $(0,0)$, $(1,0)$, and $(0,1)$. Depending on which of the 3 corners CGAL lists first as v_0, the two edge vectors emanating from that corner change according to $\mathbf{M}_0$, $\mathbf{M}_1$, or $\mathbf{M}_2$. Multiplying on the right by $\mathbf{M}_p$ shifts the local origin v_0 back to the 90° corner $(0,0)$.

By placing the target reference matrix $\mathbf{F}_{\text{target}}$ at the **midpoint** ($F_{12} = 1/4$) of the fundamental domain between the square and triangular wells,

$$\mathbf{F}_{\text{target}} = \mathbf{R}(\theta_0) \begin{pmatrix} 1 & 1/4 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{R}(\theta_0) = \begin{pmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{pmatrix}, \tag{11}$$

both the square phase ($F_{12} = 0$) and the triangular phase ($F_{12} = +1/2$) are at distance $\approx 1/4$ from $\mathbf{F}_{\text{target}}$, while the reflected obtuse triangular orientation ($F_{12} = -1/2$) and unaligned square orientations ($F_{12} = \pm 1$) are at distance $\geq 3/4$.

Combining the 3 cyclic triangle vertex permutations $\mathbf{M}_p$ ($p \in \{0,1,2\}$) with the 4 discrete 90° reference rotations $\mathbf{R}_r \in SL(2,\mathbb{Z})$,

$$\mathbf{R}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{R}_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{R}_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{R}_3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{12}$$

yields the single formula used for all elements:

$$\boxed{\mathbf{F}_{\text{aligned}} = \underset{\substack{p\in\{0,1,2\} \\ r\in\{0,1,2,3\}}}{\arg\min} \left\| \mathbf{F}\,\mathbf{M}_p\,\mathbf{R}_r^{\mathsf{T}} - \mathbf{F}_{\text{target}} \right\|_F^2} \tag{13}$$

where $\|\mathbf{A}\|_F^2 = \operatorname{tr}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \sum_{i,j} A_{ij}^2$ is the squared Frobenius norm (`.squaredNorm()` in Eigen).

3 Why Nothing Downstream Needs to Change Once $\mathbf{F}_{\text{aligned}}$ is Found

Drop-In Replacement Property.
Because $\mathbf{F}_{\text{aligned}} = \mathbf{F}\mathbf{N}$ with $\mathbf{N} \in SL(2,\mathbb{Z})$ ($\det \mathbf{N} = +1$), $\mathbf{F}_{\text{aligned}}$ belongs to the **exact same $GL(2,\mathbb{Z})$ lattice equivalence class** as $\mathbf{F}$. Therefore, replacing $\mathbf{F}$ by $\mathbf{F}_{\text{aligned}}$ at the very start of the stress routine leaves the reduced metric $\mathbf{C}_{\text{red}}$, the strain energy W, and the Cauchy stress $\boldsymbol{\sigma}$ mathematically unchanged, while automatically aligning the First Piola–Kirchhoff stress $\mathbf{P}_{\text{aligned}}$ and Projected Stress τ_{aligned}.

Once $\mathbf{F}_{\text{aligned}}$ is computed via Eq. (13), the existing stress pipeline proceeds verbatim:

1. **Aligned Right Cauchy–Green Metric Tensor**:

$$\mathbf{C}_{\text{aligned}} = \mathbf{F}_{\text{aligned}}^{\mathsf{T}}\mathbf{F}_{\text{aligned}} = \mathbf{N}^{\mathsf{T}}\mathbf{F}^{\mathsf{T}}\mathbf{F}\mathbf{N} = \mathbf{N}^{\mathsf{T}}\mathbf{C}\mathbf{N}. \tag{14}$$

2. **Lagrange Reduction to the Fundamental Domain**: Applying `lagrange::reduce`($\mathbf{C}_{\text{aligned}}$) finds the unimodular reduction matrix $\mathbf{M} \in GL(2,\mathbb{Z})$ that maps $\mathbf{C}_{\text{aligned}}$ into the unique Lagrange-reduced metric $\mathbf{C}_{\text{red}}$:

$$\mathbf{C}_{\text{red}} = \mathbf{M}^{\mathsf{T}}\mathbf{C}_{\text{aligned}}\mathbf{M}. \tag{15}$$

   For bulk elastic elements, $\mathbf{C}_{\text{aligned}}$ is already in the fundamental domain, so $\mathbf{M} = \mathbf{I}$. For plastically slipped elements (or elements at dislocation cores), $\mathbf{M}$ automatically strips the plastic lattice shear!

3. **Constitutive Energy Derivative**:

$$\mathbf{S}_{\text{red}} = \frac{1}{A_{\text{cell}}}\frac{\partial W(\mathbf{C}_{\text{red}})}{\partial \mathbf{C}_{\text{red}}}. \tag{16}$$

4. **Aligned First Piola–Kirchhoff Stress Tensor**:

$$\boxed{\mathbf{P}_{\text{aligned}} = 2\,\mathbf{F}_{\text{aligned}}\,\mathbf{M}\,\mathbf{S}_{\text{red}}\,\mathbf{M}^{\mathsf{T}}.} \tag{17}$$

5. **Cauchy Stress Tensor (Identical to Unaligned $\boldsymbol{\sigma}$)**:

$$\boxed{\boldsymbol{\sigma}_{\text{aligned}} = \frac{1}{\det \mathbf{F}_{\text{aligned}}}\mathbf{P}_{\text{aligned}}\mathbf{F}_{\text{aligned}}^{\mathsf{T}} = \frac{2}{\det \mathbf{F}_{\text{aligned}}}(\mathbf{F}_{\text{aligned}}\mathbf{M})\,\mathbf{S}_{\text{red}}\,(\mathbf{F}_{\text{aligned}}\mathbf{M})^{\mathsf{T}} = \boldsymbol{\sigma}.} \tag{18}$$

   Proof of invariance: Since $\mathbf{C}_{\text{aligned}} = \mathbf{N}^{\mathsf{T}}\mathbf{C}\mathbf{N}$ and $\det \mathbf{N} = 1$, both $\mathbf{C}$ and $\mathbf{C}_{\text{aligned}}$ reduce to the same canonical metric $\mathbf{C}_{\text{red}}$ in the fundamental domain, with $\mathbf{F}_{\text{aligned}}\mathbf{M} = \mathbf{F}\mathbf{M}_{\text{orig}}$ (up to lattice point-group symmetries of W). Hence $(\mathbf{F}_{\text{aligned}}\mathbf{M})\mathbf{S}_{\text{red}}(\mathbf{F}_{\text{aligned}}\mathbf{M})^{\mathsf{T}}$ and $\det \mathbf{F}_{\text{aligned}} = \det \mathbf{F} \det \mathbf{N} = \det \mathbf{F}$ are completely invariant.

6. **Lab-Frame First Piola–Kirchhoff and Projected Stress**: When the crystal is rotated by θ_0 (rotate-and-cut CSL or fixed-BC supercell) and sheared in the lab frame with $\mathbf{F}_{\text{ext}}(\alpha) = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$, rotating the reference index of $\mathbf{P}_{\text{aligned}}$ by $\mathbf{R}(\theta_0)^{\mathsf{T}}$ gives the lab-frame First Piola–Kirchhoff tensor and the work-conjugate projected stress:

$$\mathbf{P}_{\text{lab}} = \mathbf{P}_{\text{aligned}}\,\mathbf{R}(\theta_0)^{\mathsf{T}}, \qquad \boxed{\sigma_{\text{proj}} = \mathbf{P}_{\text{lab}} : \frac{\partial \mathbf{F}_{\text{ext}}}{\partial \alpha} = (\mathbf{P}_{\text{lab}})_{12} = P_{11}^{\text{aligned}} \sin\theta_0 + P_{12}^{\text{aligned}} \cos\theta_0.} \tag{19}$$

4 Worked Numerical Examples from Simulation

The tables below show actual element tensors extracted from a 100×100 square-lattice simulation (`checkpoint_00007.chk` at external stretch $\alpha = 0.1419$).

4.1 Example 1: Alternating Bulk Elements (`Element 0` vs. `Element 1`)

`Element 0` is a lower-left triangle ($\triangle$) whose CGAL vertex order shifted its columns by $\mathbf{M}_2$, while `Element 1` is the adjacent upper-right triangle ($\triangledown$) with opposite orientation:

Quantity	Element 0 (Bulk △)	Element 1 (Bulk ▽)
Raw $\mathbf{F}$	$\begin{pmatrix} -1.15035 & -0.00011 \\ 0.00010 & 0.86927 \end{pmatrix}$	$\begin{pmatrix} -1.15039 & 1.15024 \\ 0.00007 & -0.86937 \end{pmatrix}$
Aligned $\mathbf{F}_{\text{aligned}}$	$\begin{pmatrix} 1.15024 & -0.00011 \\ -0.00010 & 0.86927 \end{pmatrix}$	$\begin{pmatrix} 1.15024 & -0.00015 \\ -0.00007 & 0.86930 \end{pmatrix}$
Raw $\mathbf{P}$	$\begin{pmatrix} +0.39109 & -0.00013 \\ +0.00037 & -0.50618 \end{pmatrix}$	$\begin{pmatrix} +0.39137 & -0.00004 \\ -0.00009 & +\mathbf{0.50583} \end{pmatrix}$
Aligned $\mathbf{P}_{\text{aligned}}$	$\begin{pmatrix} -0.39109 & -0.00013 \\ -0.00037 & -\mathbf{0.50618} \end{pmatrix}$	$\begin{pmatrix} -0.39137 & +0.00004 \\ +0.00009 & -\mathbf{0.50583} \end{pmatrix}$
Raw Cauchy $\boldsymbol{\sigma}$	$\begin{pmatrix} -0.44985 & -0.00036 \\ -0.00036 & -0.44001 \end{pmatrix}$	$\begin{pmatrix} -0.45017 & +0.00010 \\ +0.00010 & -0.43972 \end{pmatrix}$
Aligned Cauchy $\boldsymbol{\sigma}_{\text{aligned}}$	$\begin{pmatrix} -0.44985 & -0.00036 \\ -0.00036 & -0.44001 \end{pmatrix}$	$\begin{pmatrix} -0.45017 & +0.00010 \\ +0.00010 & -0.43972 \end{pmatrix}$

Notice two key facts in this table:

- In the raw calculation, P_{22} jumps from -0.50618 on `Element 0` to $+0.50583$ on `Element 1`, and P_{11} has the wrong sign ($+0.391$) relative to σ_{11} (-0.450).
- With $\mathbf{F}_{\text{aligned}}$, both elements have $\mathbf{F}_{\text{aligned}} \approx \text{diag}(1.150, 0.869)$, $\mathbf{P}_{\text{aligned}} \approx \text{diag}(-0.391, -0.506)$ with consistent sign across the entire mesh, and $\boldsymbol{\sigma}_{\text{aligned}} \equiv \boldsymbol{\sigma}$ to machine precision.

4.2 Example 2: Plastically Slipped Elements (`Element 16328` and `Element 16126`)

For triangles that have undergone a plastic slip or sit next to a dislocation core, $\mathbf{F}_{\text{aligned}}$ removes the arbitrary CGAL vertex permutation while Lagrange reduction ($\mathbf{M}$ in Eq. (17)) reduces the slipped metric back to the fundamental elastic domain:

Quantity	Element 16328 (Slipped)	Element 16126 (Slipped)
Raw $\mathbf{F}$	$\begin{pmatrix} -1.14893 & 2.29950 \\ -0.00297 & -0.87059 \end{pmatrix}$	$\begin{pmatrix} 1.15165 & 1.14974 \\ 0.87404 & -0.00331 \end{pmatrix}$
Aligned $\mathbf{F}_{\text{aligned}}$	$\begin{pmatrix} 1.15057 & -1.14893 \\ -0.87355 & -0.00297 \end{pmatrix}$	$\begin{pmatrix} 1.14974 & -1.15165 \\ -0.00331 & -0.87404 \end{pmatrix}$
Raw $\mathbf{P}$	$\begin{pmatrix} +0.38651 & -0.02527 \\ +0.02731 & +0.49104 \end{pmatrix}$	$\begin{pmatrix} -0.02266 & -0.37120 \\ -0.50378 & +0.02636 \end{pmatrix}$
Aligned $\mathbf{P}_{\text{aligned}}$	$\begin{pmatrix} -0.36124 & -0.38651 \\ +0.51835 & -0.02731 \end{pmatrix}$	$\begin{pmatrix} -0.37120 & +0.02266 \\ +0.02636 & +0.50378 \end{pmatrix}$
Raw Cauchy $\boldsymbol{\sigma}$	$\begin{pmatrix} -0.41502 & +0.03097 \\ +0.03097 & -0.45353 \end{pmatrix}$	$\begin{pmatrix} -0.42703 & -0.02985 \\ -0.02985 & -0.44042 \end{pmatrix}$
Aligned Cauchy $\boldsymbol{\sigma}_{\text{aligned}}$	$\begin{pmatrix} -0.41502 & +0.03097 \\ +0.03097 & -0.45353 \end{pmatrix}$	$\begin{pmatrix} -0.42703 & -0.02985 \\ -0.02985 & -0.44042 \end{pmatrix}$